

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : Experiments (High)
Assessment Tool Weightage: 40%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	For the relation STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.	BL2 – Understand	5
2	Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.	BL3 – Apply	5
3	Given relation R(A,B,C,D,E) with FDs: $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$. Analyze and decompose it to 2NF eliminating partial dependencies.	BL3 – Apply	5
4	Apply 3NF decomposition on the relation EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) with transitive dependencies.	BL3 – Apply	5
5	Verify whether R(A,B,C,D) with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ is in BCNF. If not, decompose it into BCNF.	BL3 – Apply	5
6	Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.	BL3 – Apply	5
7	Identify the multi-valued dependencies in TEACHER(TID, Subject, Hobby) and decompose it into 4NF.	BL3 – Apply	5

8	Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).	BL3 – Apply	5
9	Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.	BL2 – Understand	5
10	Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.	BL3 – Apply	5

Code of Conduct:

I _ R . M o n i k a (1 9 2 3 2 4 2 8 1) _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
R.Monika

RUBRICS FOR CALCULATION:

Criteria	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Q1. STUDENT_COURSE – Functional Dependencies and Candidate Keys

Given relation

STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade)

Assume:

- A student ID uniquely identifies a student.
- A course ID uniquely identifies a course.
- A student's grade depends on both the student and course.

Functional Dependencies

The functional dependencies are:

1. $SID \rightarrow SName$
2. $CID \rightarrow CName$
3. $CID \rightarrow Instructor$
4. $(SID, CID) \rightarrow Grade$

Therefore,

$F = \{ SID \rightarrow SName, CID \rightarrow CName, CID \rightarrow Instructor, SID, CID \rightarrow Grade \}$

Finding Candidate Key

Take:

$(SID, CID)^+$

Start:

$\{SID, CID\}$

Using $SID \rightarrow SName$:

$\{SID, CID, SName\}$

Using $CID \rightarrow CName, Instructor$:

$\{SID, CID, SName, CName, Instructor\}$

Using $SID, CID \rightarrow Grade$:

$\{SID, CID, SName, CName, Instructor, Grade\}$

Therefore,

$(SID, CID)^+ = \text{all attributes}$

Hence:

Candidate Key = $\{SID, CID\}$

Neither SID nor CID alone can determine all attributes.

Final Answer:

Candidate Key = (SID, CID)

Q2. Convert an Unnormalized Relation into 1NF

Example Unnormalized Relation

Consider:

STUDENT(SID, SName, PhoneNumbers)

SID	SName	PhoneNumbers
S01	Ravi	"98768765"
S02	Priya	"91238234"

Here, PhoneNumbers contains multiple values.

Therefore, it violates **1NF**.

Step 1: Identify repeating group

For S01:

PhoneNumbers = {9876, 8765}

For S02:

PhoneNumbers = {9123, 8234}

A 1NF relation must contain **atomic values**.

Step 2: Convert to 1NF

Create one row for each phone number.

SID	SName	PhoneNumber
S01	Ravi	9876
S01	Ravi	8765
S02	Priya	9123
S02	Priya	8234

Now every attribute contains a **single atomic value**.

Primary Key

The combination:

(SID, PhoneNumber)

can uniquely identify each record.

Conclusion

The original relation was not in 1NF because PhoneNumbers contained multiple values.

After decomposition:

STUDENT_PHONE(SID, SName, PhoneNumber)

is in **1NF**.

Q3. Decompose into 2NF

Given

Relation: $R(A, B, C, D, E)$

Functional Dependencies:

- $A \rightarrow BC$
- $BC \rightarrow D$
- $D \rightarrow E$

We need to decompose into 2NF.

Step 1: Find Candidate Key

Calculate A^+ :

$A^+ = \{A\}$

Using $A \rightarrow BC$: $= \{A, B, C\}$

Using $BC \rightarrow D$: $= \{A, B, C, D\}$

Using $D \rightarrow E$: $= \{A, B, C, D, E\}$

Therefore:

$A^+ = \{A, B, C, D, E\}$

So: **Candidate Key** = **A**

Since the candidate key consists of only one attribute, there cannot be any partial dependency on a part of the key.

Therefore, the relation is already in 2NF.

Step 2: Check dependencies

$A \rightarrow BC$ is a dependency from the complete key.

$BC \rightarrow D$ and $D \rightarrow E$ do not violate 2NF because 2NF is concerned with **partial dependency on a proper subset of a composite candidate key**.

Conclusion

Since the candidate key is a single attribute:

$R(A, B, C, D, E)$ is in 2NF.

However, it has a **transitive dependency**:

$A \rightarrow BC \rightarrow D \rightarrow E$

Therefore, it may require decomposition to achieve 3NF

Q4. 3NF Decomposition of EMPLOYEE

Given relation

EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID)

Functional dependencies:

1. **EmpID** → **EmpName, DeptID**
2. **DeptID** → **DeptName, ManagerID**

Step 1: Identify Candidate Key

EmpID determines:

EmpID → EmpName, DeptID

DeptID → DeptName, ManagerID

Therefore:

EmpID+ = {EmpID, EmpName, DeptID, DeptName, ManagerID}

EmpID	EmpName	DeptID	DeptName	ManagerID
E01	Ravi	D01	Computer Science	M01
E02	Priya	D01	Computer Science	M01
E03	Arun	D02	Information Technology	M02

Hence:

Candidate Key = EmpID

Step 2: Identify transitive dependency

We have:

EmpID → **DeptID** and

DeptID → **DeptName, ManagerID**

EmpID	EmpName	DeptID	DeptName	ManagerID
E01	Ravi	D01	Computer Science	M01
E02	Priya	D01	Computer Science	M01
E03	Arun	D02	Information Technology	M02

Therefore:

EmpID → **DeptID** → **DeptName, ManagerID**

This is a **transitive dependency**.

Hence the relation is not in 3NF.

Step 3: Decompose

Create:

Relation 1

EMPLOYEE(EmpID, EmpName, DeptID)

EmpID	EmpName	DeptID
E01	Ravi	D01
E02	Priya	D01
E03	Arun	D02

FD: **EmpID** → **EmpName, DeptID**

Candidate Key: **EmpID**

Relation 2

DEPARTMENT(DeptID, DeptName, ManagerID)

DeptID	DeptName	ManagerID
D01	Computer Science	M01
D02	Information Technology	M02

FD: **DeptID** → **DeptName, ManagerID**

Candidate Key: **DeptID**

Final decomposition

EMPLOYEE(EmpID, EmpName, DeptID)

DEPARTMENT(DeptID, DeptName, ManagerID)

Why is it in 3NF?

In EMPLOYEE: **EmpID** → **EmpName, DeptID**

EmpID is a candidate key.

In DEPARTMENT: **DeptID** → **DeptName, ManagerID**

DeptID is a candidate key.

Therefore, both relations satisfy 3NF.

Q5. Check BCNF and Decompose

Given

Relation:

R(A, B, C, D)

A	B	C	D
a1	b1	c1	d1
a2	b2	c2	d2
a3	b3	c3	d3

Functional dependencies:

- **$AB \rightarrow C$**
- **$C \rightarrow D$**
- **$D \rightarrow A$**

We need to determine whether R is in BCNF.

Step 1: Find Candidate Keys

AB+

$AB^+ = \{A, B\}$

Using $AB \rightarrow C$: $= \{A, B, C\}$

Using $C \rightarrow D$: $= \{A, B, C, D\}$

Therefore:

AB is a candidate key.

BC+

$BC^+ = \{B, C\}$

Using $C \rightarrow D$: $= \{B, C, D\}$

Using $D \rightarrow A$: $= \{A, B, C, D\}$

Therefore: **BC is also a candidate key.**

So candidate keys are: **AB and BC**

Step 2: Check BCNF

BCNF condition: For every non-trivial FD $X \rightarrow Y$, X must be a superkey.

Check:

$AB \rightarrow C$

AB is a candidate key.

Therefore, satisfied.

$C \rightarrow D$

C is **not** a superkey because:

$C^+ = \{C, D, A\}$

It does not contain B.

Therefore:

$C \rightarrow D$ violates BCNF.

$D \rightarrow A$

D is also not a superkey.

$D^+ = \{D, A\}$

It does not contain B or C.

Therefore: **$D \rightarrow A$ also violates BCNF.**

Hence: **R is NOT in BCNF.**

Step 3: BCNF Decomposition

Using $C \rightarrow D$:

Create: **R1(C,D)** and **R2(A,B,C)**

R1(C,D)

C	D
c1	d1
c2	d2
c3	d3

FD: $C \rightarrow D$

C is the key of R1.

Therefore **R1 is in BCNF.**

R2(A,B,C)

A	B	C
a1	b1	c1
a2	b2	c2
a3	b3	c3

FD: $AB \rightarrow C$

AB is a key.

Therefore **R2 is in BCNF.**

Q6. Lossless-Join Decomposition Using Chase Algorithm

Since the question does not specify a particular relation, consider this standard example.

Given

Relation: **R(A, B, C)**

Functional dependency: **$A \rightarrow B$**

Decompose into:

R1(A,B)

R2(A,C)

We need to verify whether the decomposition is **lossless**.

Step 1: Create Chase Table

Create columns A, B and C.

For R1(A,B), use:

Row	A	B	C
R1	a	b	c1
R2	a	b1	c

Here:

- Common attribute = A
- R1 has A and B
- R2 has A and C

Step 2: Apply FD $A \rightarrow B$

Both rows contain the same value for A: **$A = a$**

Therefore, according to: **$A \rightarrow B$** their B values must become the same.

Row	A	B	C
R1	a	b	c1
R2	a	b	c

Step 3: Check for a Distinguished Row

The first row now contains: **a, b, c**

where all attributes have the common/distinguished symbols.

Therefore, the chase succeeds.

Result

Since a row becomes completely distinguished: **The decomposition is lossless-join.**

Conclusion

$R(A,B,C) \rightarrow R1(A,B), R2(A,C)$ is a lossless decomposition.

Q7. MVD and 4NF Decomposition

Given

Relation: **TEACHER(TID, Subject, Hobby)**

Suppose a teacher can teach multiple subjects and can have multiple hobbies independently.

Example:

TID	Subject	Hobby
T01	DBMS	Cricket
T01	DBMS	Music
T01	AI	Cricket
T01	AI	Music

Step 1: Identify Multivalued Dependencies

Subjects and hobbies are independent. Therefore:

TID $\twoheadrightarrow$ **Subject** and

TID $\twoheadrightarrow$ **Hobby**

These are **multivalued dependencies (MVDs)**.

Step 2: Check 4NF

A relation is in 4NF if: For every non-trivial MVD $X \twoheadrightarrow Y$, X must be a superkey.

Here: **TID** $\twoheadrightarrow$ **Subject**

But TID alone does not determine all attributes. Therefore, TID is not a superkey.

So the relation violates **4NF**.

Step 3: Decompose

Create two relations: **1.TEACHER_SUBJECT**

TID	Subject
T01	DBMS
T01	AI

2. TEACHER_HOBBY

TID	Hobby
T01	Cricket
T01	Music

Final Answer

TEACHER_SUBJECT(TID, Subject)

TEACHER_HOBBY(TID, Hobby)

Both relations are in **4NF** because their determinants are keys.

Q8. Join Dependency and 5NF

Given

Consider: SUPPLY(Supplier, Part, Project)

Suppose the information can be represented independently through:

- Supplier supplies a Part
- Supplier works for a Project
- Part is used in a Project

This creates a **join dependency**.

Step 1: Join Dependency

The relation can be decomposed into:

SP(Supplier, Part)

SJ(Supplier, Project)

PJ(Part, Project)

The join dependency is:

$$(Supplier, Part, Project) \rightarrow SP \bowtie SJ \bowtie PJ$$

Step 2: Why redundancy occurs?

Suppose:

Supplier	Part	Project
S1	P1	J1
S1	P1	J2
S1	P2	J1
S1	P2	J2

If the three facts are independent, storing every combination creates redundant tuples.

For example,

information about S1 supplying P1 is repeated for every project.

Step 3: Decompose

Create: **SP**

SP(Supplier, Part)

Supplier	Part
S1	P1
S1	P2

SJ

SJ(Supplier, Project)

Supplier	Project
S1	J1
S1	J2

PJ

PJ(Part, Project)

Part	Project
P1	J1
P1	J2
P2	J1
P2	J2

Step 4: 5NF

A relation is in **5NF** if every non-trivial join dependency is implied by candidate keys.

The decomposition separates the independent relationships.

Final Answer

SUPPLY → **SP(Supplier, Part)**

SJ(Supplier, Project)

PJ(Part, Project)

This removes join dependency-related redundancy and produces a **5NF decomposition**.

Q9. Anomalies and Normalization to 3NF

Consider the poorly designed college relation:

COLLEGE(StudentID, StudentName, CourseID, CourseName, InstructorID, InstructorName)

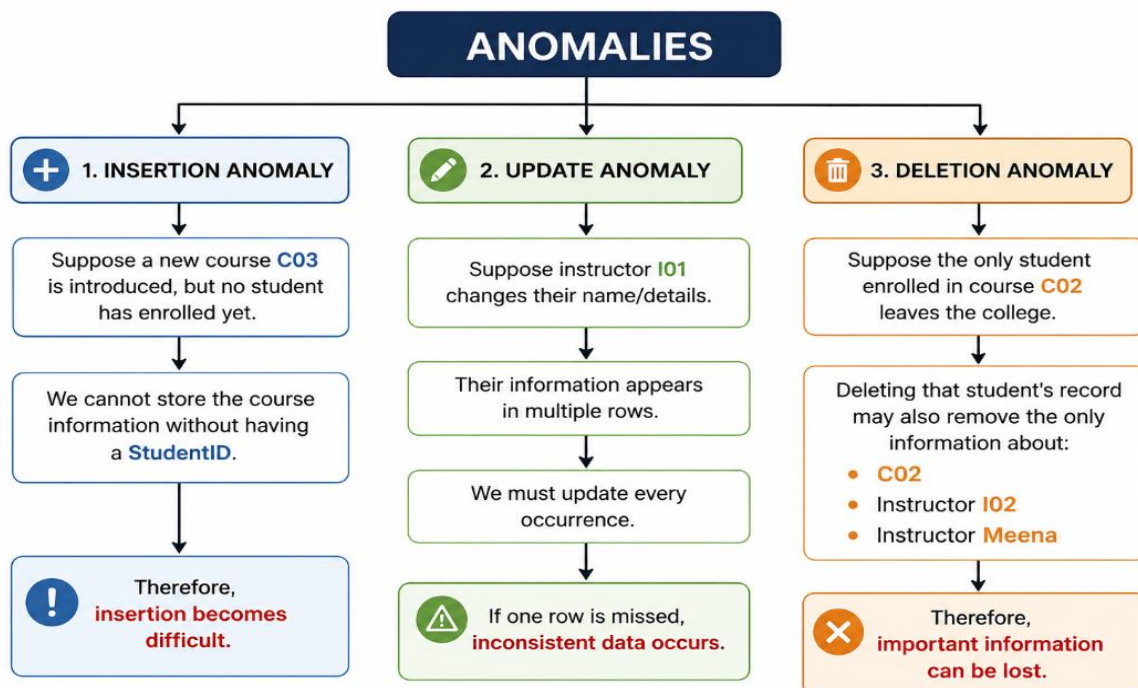
Example:

StudentID	StudentName	CourseID	CourseName	InstructorID	InstructorName
S01	Ravi	C01	DBMS	I01	Kumar
S02	Priya	C01	DBMS	I01	Kumar
S03	Arun	C02	AI	I02	Meena

Step 1: Functional Dependencies

1. StudentID $\rightarrow$ StudentName
2. CourseID $\rightarrow$ CourseName, InstructorID
3. InstructorID $\rightarrow$ InstructorName

Candidate key: (StudentID, CourseID)



Step 2: Normalize to 3NF

The transitive dependency is:

CourseID $\rightarrow$ InstructorID and

InstructorID $\rightarrow$ InstructorName

Therefore: **CourseID $\rightarrow$ InstructorID $\rightarrow$ InstructorName**

This violates 3NF.

Decomposition: STUDENT

STUDENT(StudentID, StudentName)

StudentID, StudentName

S01	Ravi
S02	Priya
S03	Arun

Primary Key: **StudentID**

COURSE

COURSE(CourseID, CourseName, InstructorID)

CourseID	CourseName	InstructorID
C01	DBMS	I01
C02	AI	I02

Primary Key: **CourseID**

INSTRUCTOR

INSTRUCTOR(InstructorID, InstructorName)

InstructorID	InstructorName
I01	Kumar
I02	Meena

Primary Key: **InstructorID**

ENROLLMENT

ENROLLMENT(StudentID, CourseID)

StudentID	CourseID
S01	C01
S02	C01
S03	C02

Primary Key: **(StudentID, CourseID)**

Final Answer

The original college relation is decomposed into:

STUDENT(StudentID, StudentName)

COURSE(CourseID, CourseName, InstructorID)

INSTRUCTOR(InstructorID, InstructorName)

ENROLLMENT(StudentID, CourseID)

This eliminates insertion, deletion and update anomalies and results in a **3NF design**.

Q10. Attribute Closure and Candidate Keys

Since the question does not specify an FD set, consider this standard example.

Given

Relation:

R(A, B, C, D, E)

Functional dependencies:

- $A \rightarrow B$
- $B \rightarrow C$
- $CD \rightarrow E$
- $E \rightarrow A$

Find attribute closures and all candidate keys.

Step 1: Calculate A^+

Start:

$A^+ = \{A\}$

Using $A \rightarrow B$:

$A^+ = \{A, B\}$

Using $B \rightarrow C$:

$A^+ = \{A, B, C\}$

We do not have D.

Therefore, $CD \rightarrow E$ cannot be applied.

So:

$A^+ = \{A, B, C\}$

A is not a key.

Step 2: Calculate E^+

Start: $E^+ = \{E\}$

Using $E \rightarrow A$: $E^+ = \{E, A\}$

Using $A \rightarrow B$: $E^+ = \{E, A, B\}$

Using $B \rightarrow C$: $E^+ = \{E, A, B, C\}$

D is missing.

Therefore: $E^+ = \{A, B, C, E\}$

E is not a key.

Step 3: Calculate CD^+

Start: $CD^+ = \{C, D\}$

Using: $CD \rightarrow E$

we get: {C,D,E}

Using: $E \rightarrow A$

we get: {A,C,D,E}

Using: $A \rightarrow B$

we get: {A,B,C,D,E}

Therefore: $CD^+ = \{A,B,C,D,E\}$

So: **CD is a candidate key.**

Step 4: Find Another Candidate Key

Consider DE^+ .

Start:

$DE^+ = \{D,E\}$

$E \rightarrow A: \{D,E,A\}$

$A \rightarrow B: \{D,E,A,B\}$

$B \rightarrow C: \{D,E,A,B,C\}$

Therefore: $DE^+ = \{A,B,C,D,E\}$

So: **DE is also a candidate key.**

Closure Summary

AttributeSet	Closure	IsCandidateKey
A	A,B,C	No
E	A,B,C,E	No
CD	A,B,C,D,E	Yes
DE	A,B,C,D,E	Yes

Final Answer

Candidate Keys = CD and DE
